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# Quantum Hall Effects in Two-Dimensional System

**Zhongqi Xiu**

**2026/04/24**

# OUTLINE

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1. Mathematical Background

2. Haldane model and Chern insulator

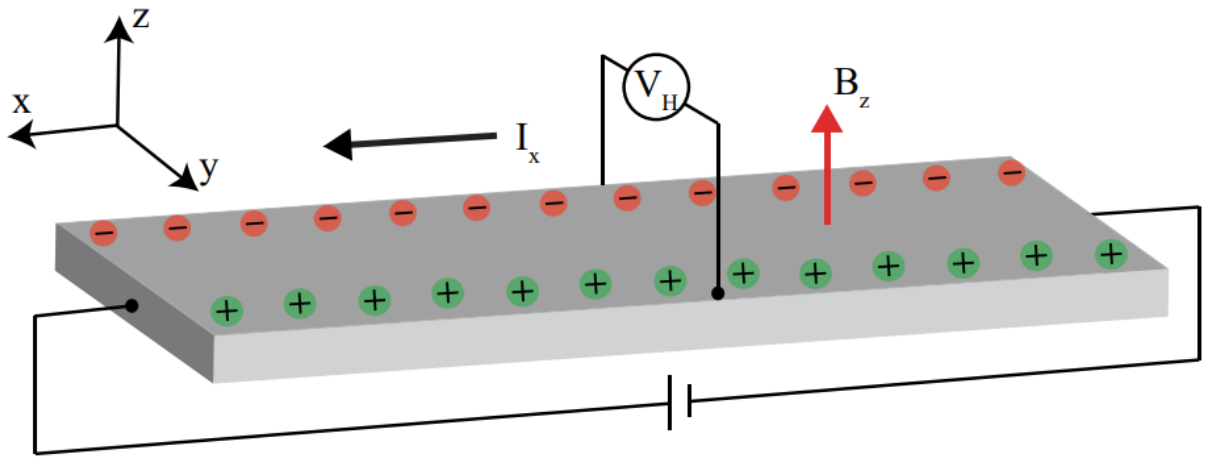
3. Optical control over Chern number

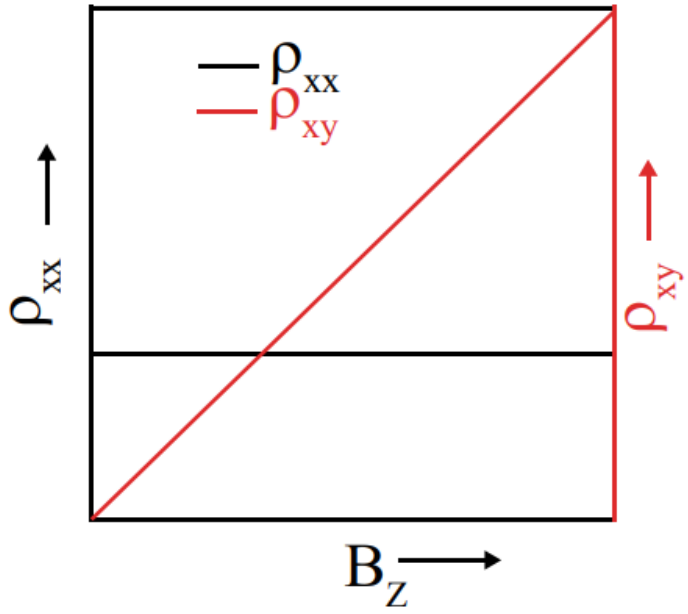
# 01.

# Mathematical Background

# Classical Hall effect

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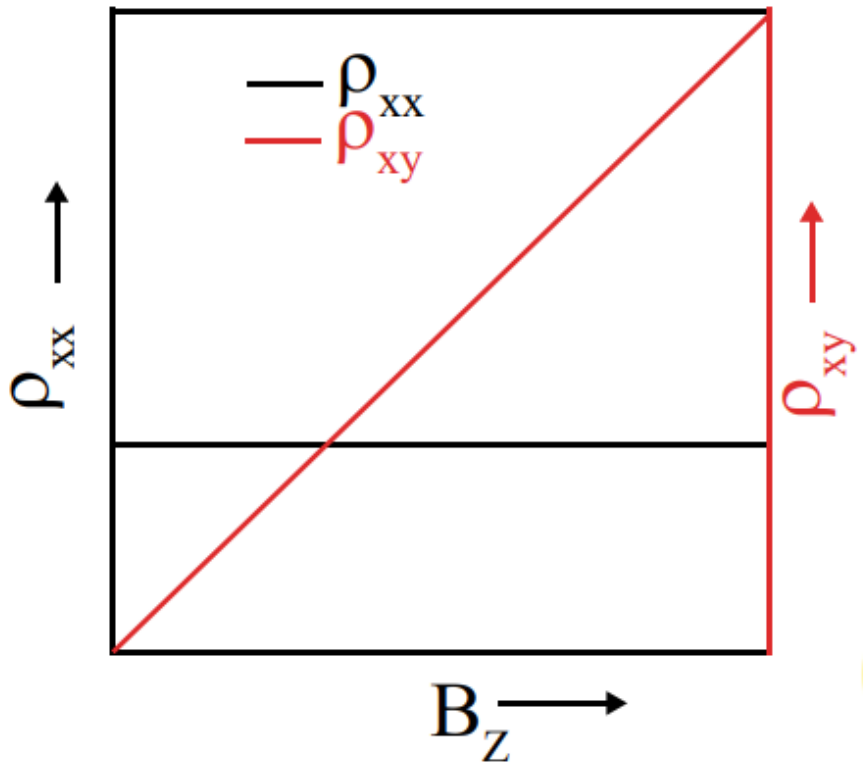




Classical Hall effect converts longitudinal current into transverse voltage

Time-reversal symmetry is broken by magnetic field (Lorentz Force) $\vec{F} = q(\vec{v} \times \vec{B})$

# Classical Hall effect

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Drude model: $\frac{d}{dt}\langle \mathbf{p}(t)\rangle = q\left(\mathbf{E} + \frac{\langle \mathbf{p}(t)\rangle}{m} \times \mathbf{B}\right) - \frac{\langle \mathbf{p}(t)\rangle}{\tau}$

$$m_e \frac{d\mathbf{v}}{dt} = -e\mathbf{E} - e(\mathbf{v} \times \mathbf{B}) - \frac{m_e \mathbf{v}}{\tau} \qquad \mathbf{j} = -ne\mathbf{v}$$

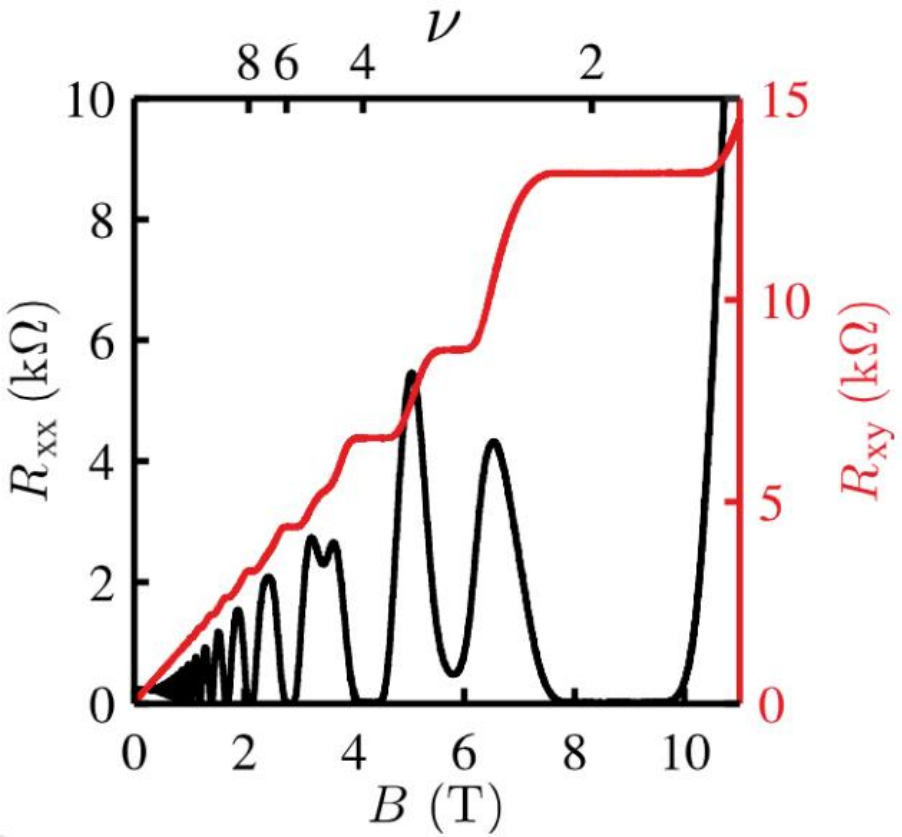
$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} m_e/ne^2\tau & B/ne \\ -B/ne & m_e/ne^2\tau \end{pmatrix} \begin{pmatrix} J_x \\ J_y \end{pmatrix}$$

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{pmatrix} \begin{pmatrix} J_x \\ J_y \end{pmatrix} \rightarrow$$

$$\rho_{xx} = \rho_{yy} = \frac{m_e}{ne^2\tau}$$

$$\rho_{xy} = -\rho_{yx} = \frac{B}{|e|n}$$

# Integer quantum Hall effect



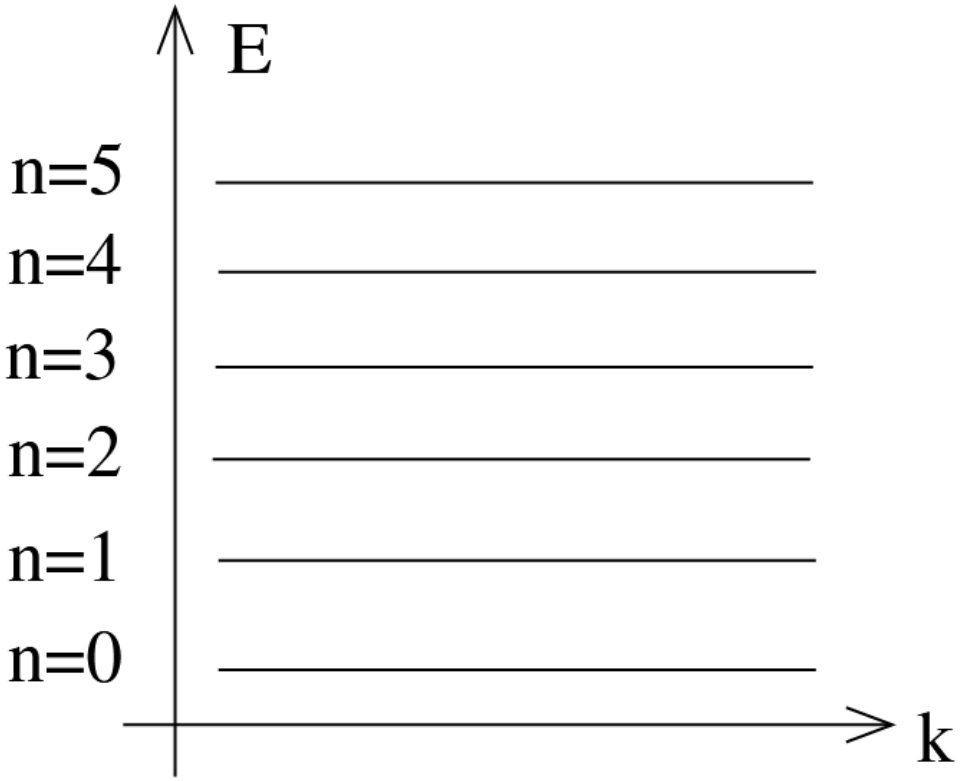

$$\hat{\mathbf{H}} = \left[\frac{1}{2m_e}((p_x - eBy)^2 + p_y^2)\right]$$

$$\left[\frac{p_y^2}{2m_e} + \frac{1}{2}m_e\omega_c^2(y - \frac{k\hbar}{eB})^2\right]\phi(y) = \epsilon\phi(y)$$

$$\epsilon_{N,k} = \left(N + \frac{1}{2}\right)\hbar\omega_c$$

New J. Phys. 14 083015

# Landau levels



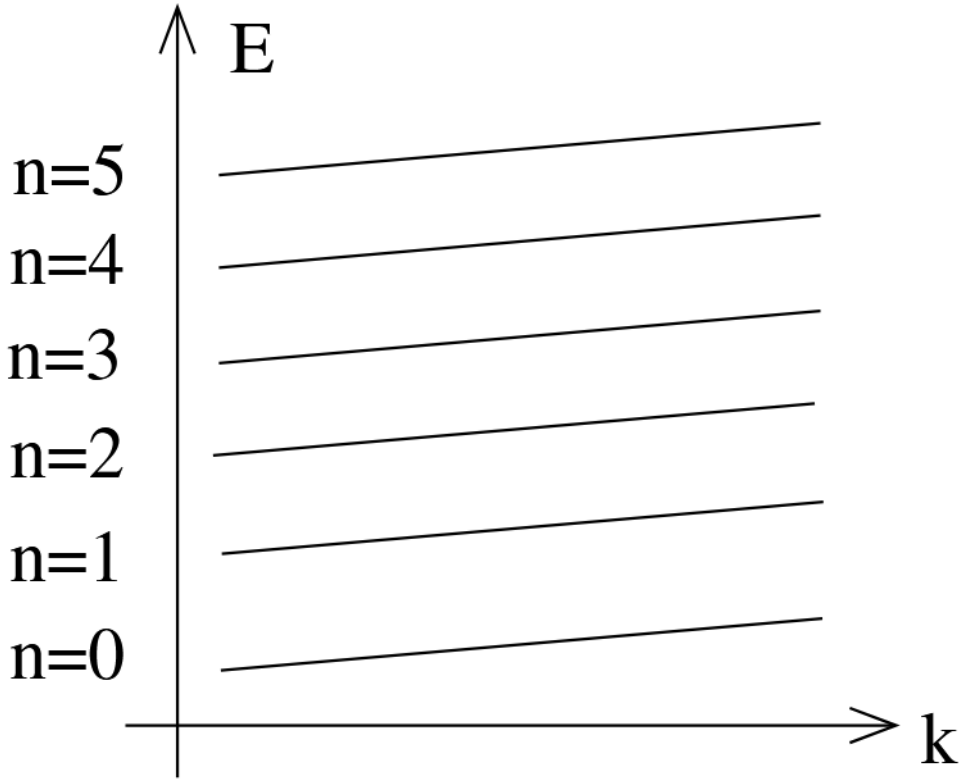

$$\epsilon_{N,k} = \left(N + \frac{1}{2}\right)\hbar\omega_c$$

$$\mathcal{N} = \frac{1}{2\pi/L_x}\int_0^{L_y/l_B^2} dk = \frac{L_x L_y}{2\pi l_B^2} = \frac{eBA}{2\pi\hbar}$$

Filling factor: $\nu = \frac{N_e}{\mathcal{N}} = \frac{n_e A}{(eBA/h)} = \frac{n_e h}{eB}$

# Landau levels with electric fields

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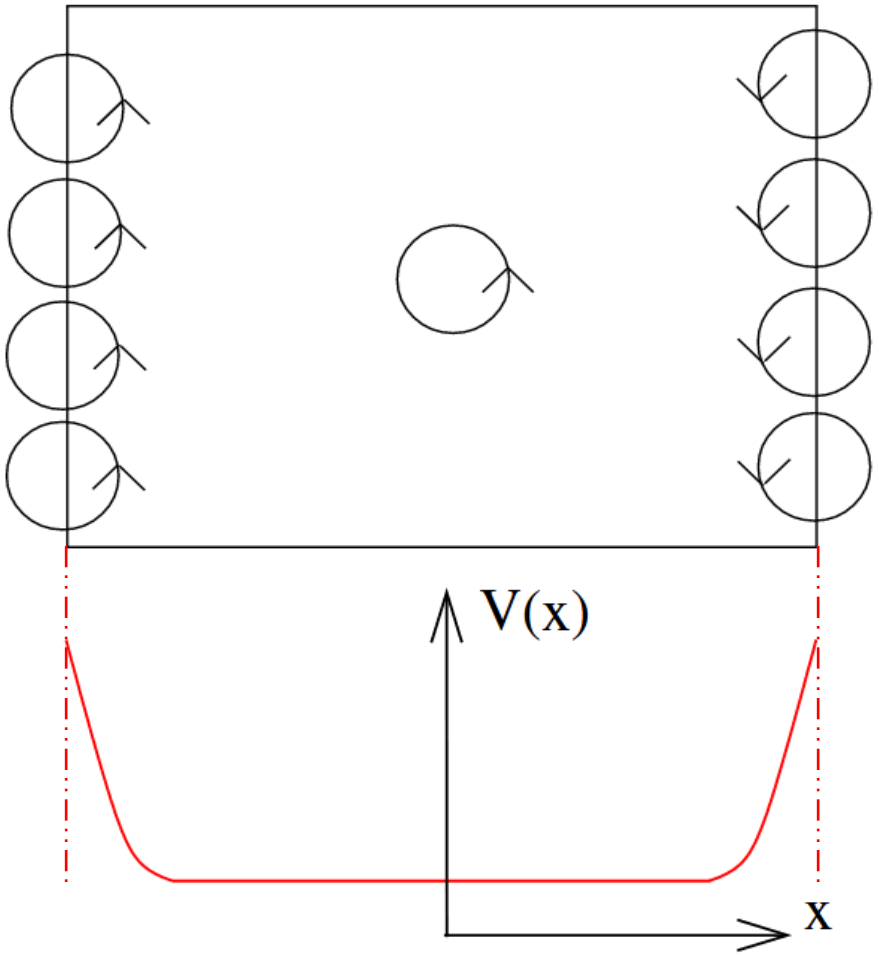



$$\left[\frac{p_y^2}{2m_e} + \frac{1}{2}m_e\omega_c^2(y - y_0)^2 - eEy\right]\phi(y) = \epsilon\phi(y)$$

$$\epsilon_{N,k} = \left(N + \frac{1}{2}\right)\hbar\omega_c + eEkl_B^2 - \frac{mE^2}{2B^2}$$

$$v_y = \frac{1}{\hbar}\frac{\partial\epsilon_{n,k}}{\partial k} = -\frac{e}{\hbar}El_B^2 = -\frac{E}{B}$$

# Edge states




$$H = \frac{1}{2m}\left(p_x^2 + (p_y + eBx)^2\right) + V(x)$$

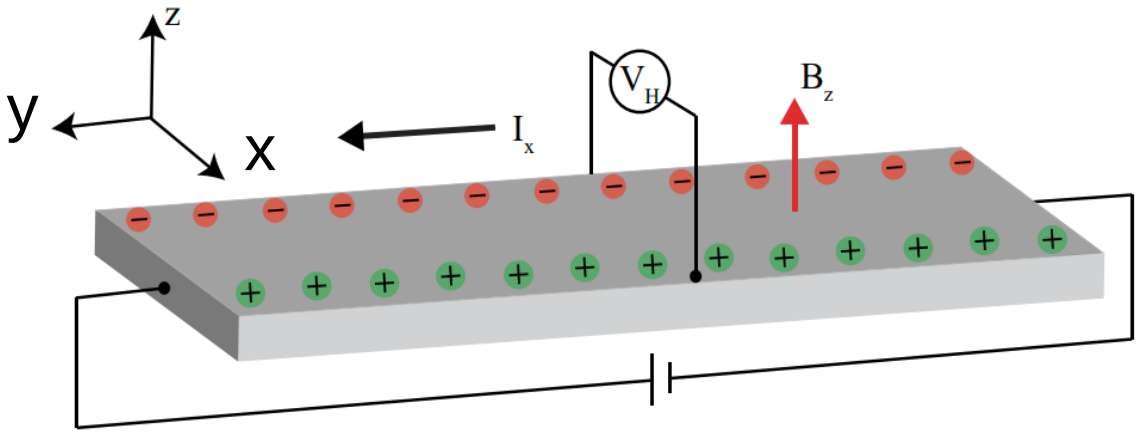
$$\epsilon_{N,k} = \left(N + \frac{1}{2}\right)\hbar\omega_c + V(kl_B^2)$$

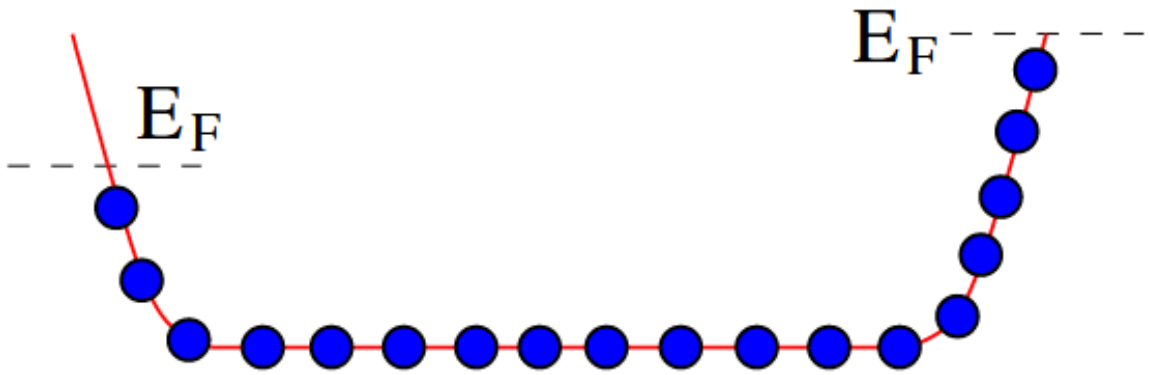
Drift velocity: $v_y(k) = \frac{1}{\hbar}\frac{\partial \epsilon_{N,k}}{\partial k} = \frac{1}{\hbar}\frac{\partial V(x_0)}{\partial k}$

$$v_y(k) = \frac{1}{\hbar}\frac{\partial V(x)}{\partial x}\Big|_{x=x_0}\frac{\partial x_0}{\partial k} = \frac{1}{eB}\frac{\partial V(x)}{\partial x}\Big|_{x=x_0}$$

# Quantized Hall conductivity

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$$\epsilon_{N,k} = \left(N + \frac{1}{2}\right)\hbar\omega_c + V(kl_B^2)$$

$$I_y = -e\int \frac{dk}{2\pi} v_y(k) = \frac{e}{2\pi l_B^2}\int dx \frac{1}{eB}\frac{\partial V}{\partial x} = -\frac{e}{2\pi\hbar}\Delta\mu$$

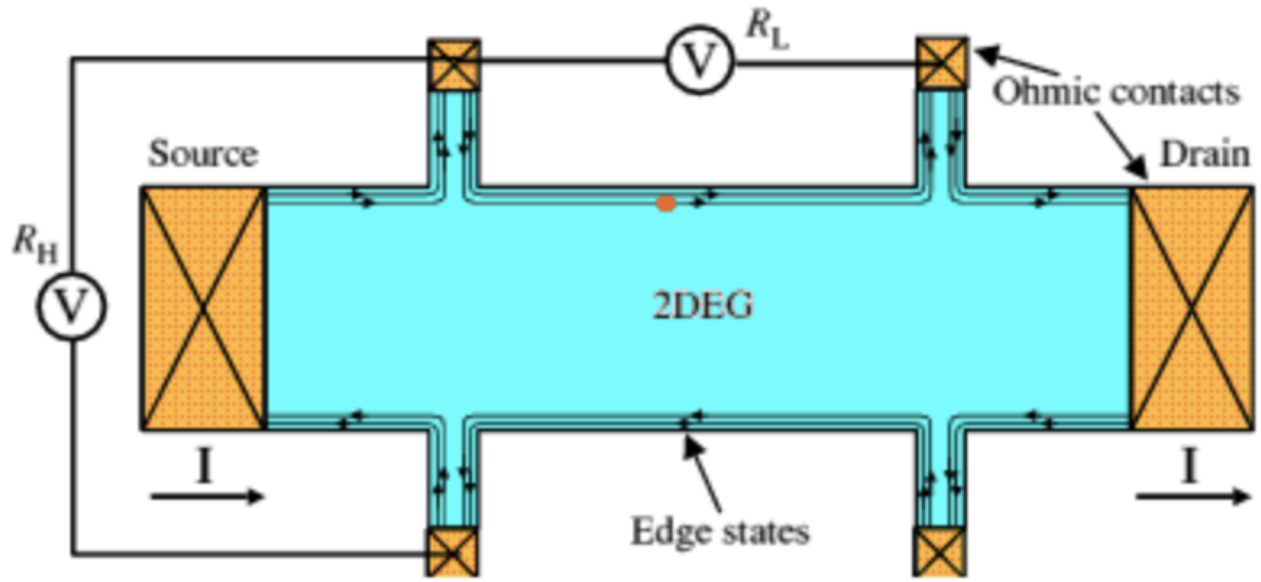
$$eV_H = \Delta\mu \qquad \sigma_{xy} = \frac{I_y}{V_H} = -\frac{e^2}{2\pi\hbar}$$

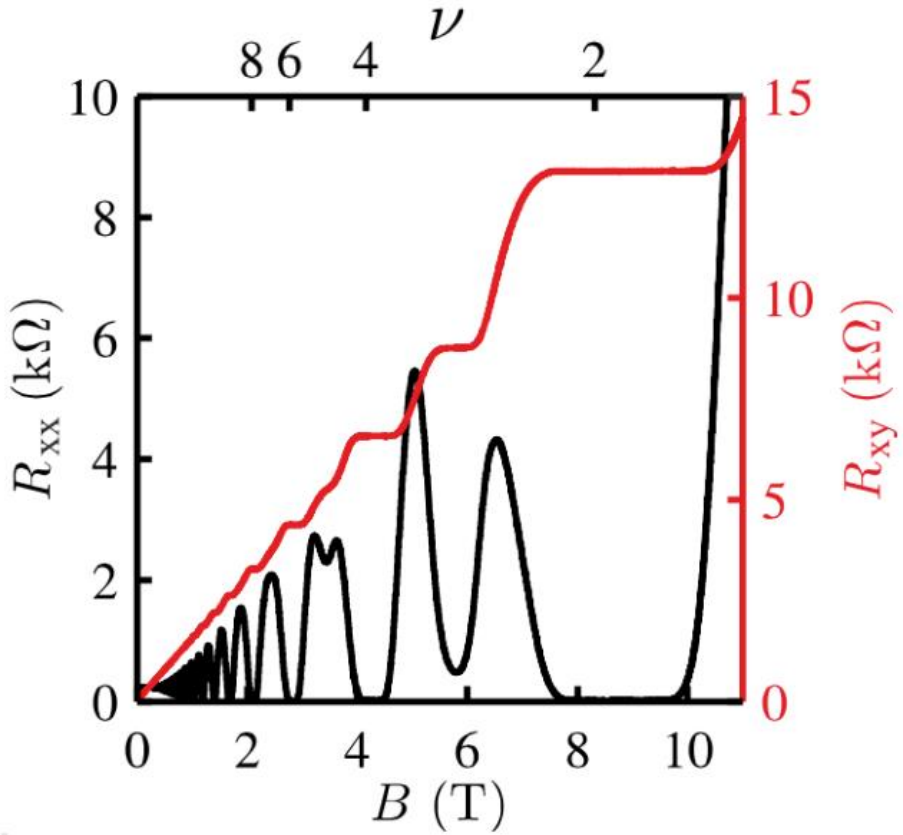
$$\sigma_{xy,sum} = \frac{\nu I_y}{V_H} = -\frac{\nu e^2}{2\pi\hbar}$$

where *v* is filling factor

# Longitudinal resistivity and Robustness

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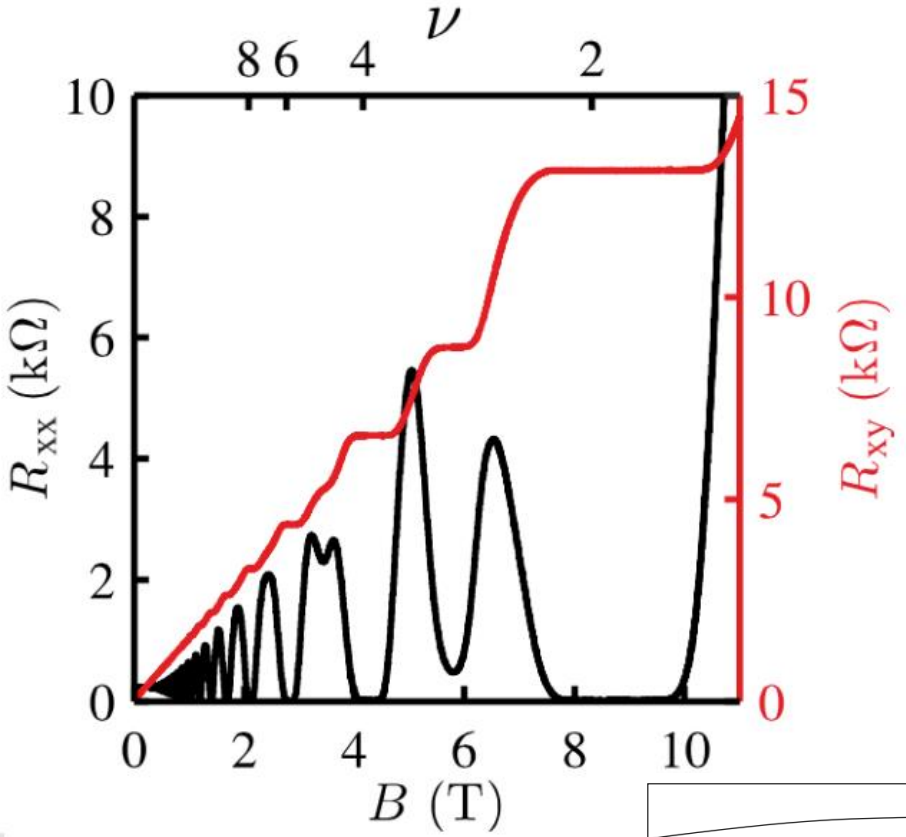


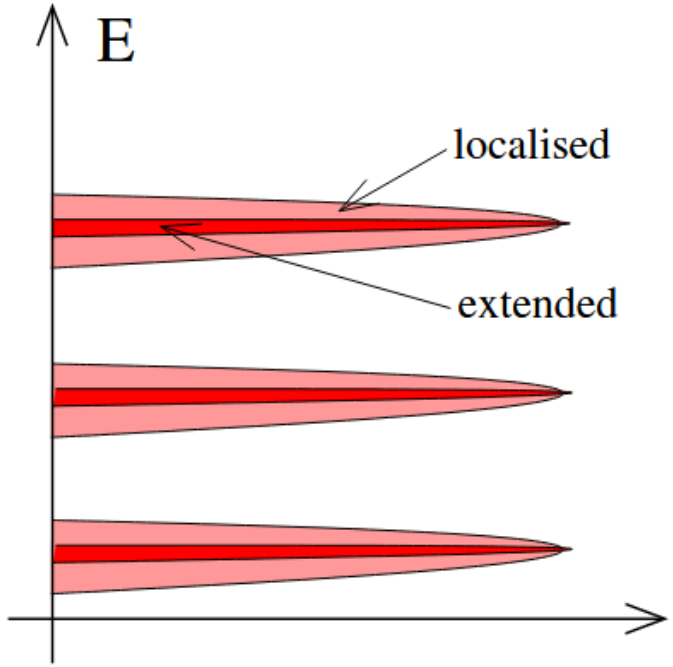



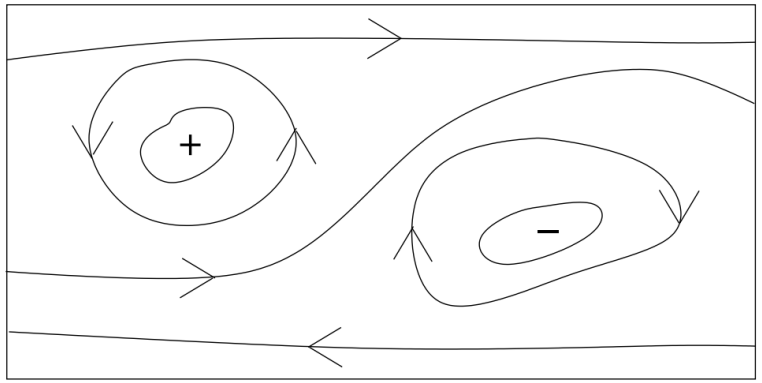

$$\sigma_{xx} = \frac{\rho_{xx}}{\rho_{xx}^2 + \rho_{xy}^2} \quad \text{and} \quad \sigma_{xy} = \frac{-\rho_{xy}}{\rho_{xx}^2 + \rho_{xy}^2}$$

# Localized and extended state

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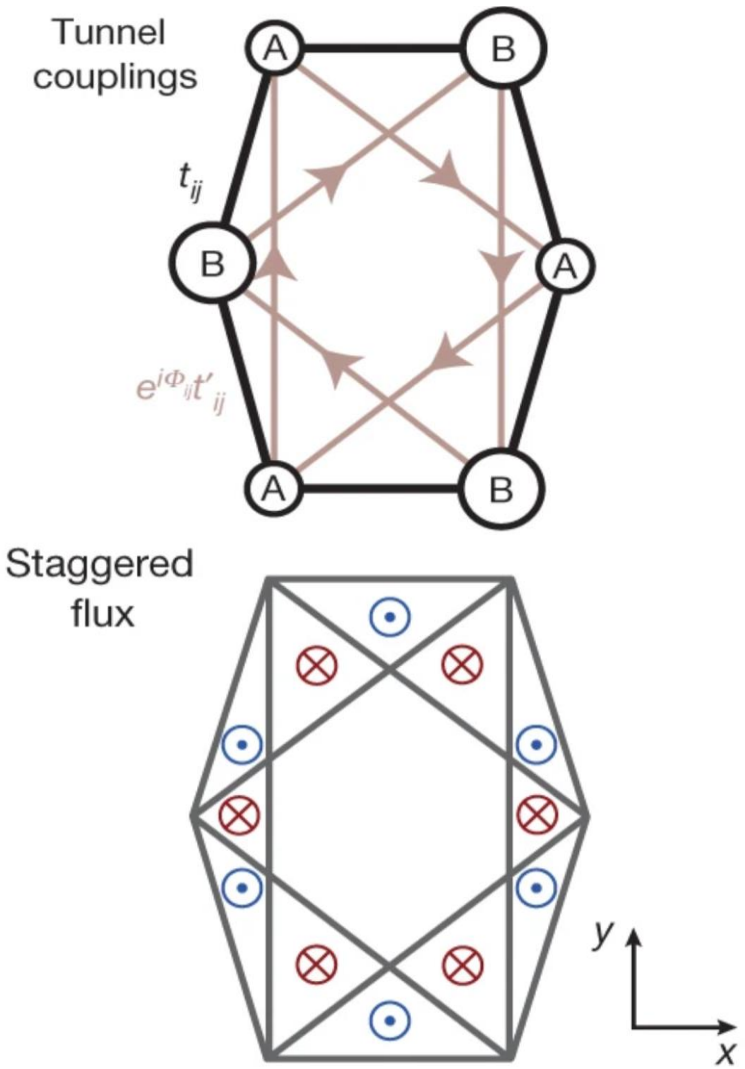



# 02.

# Haldane model and Chern insulator

# Haldane model



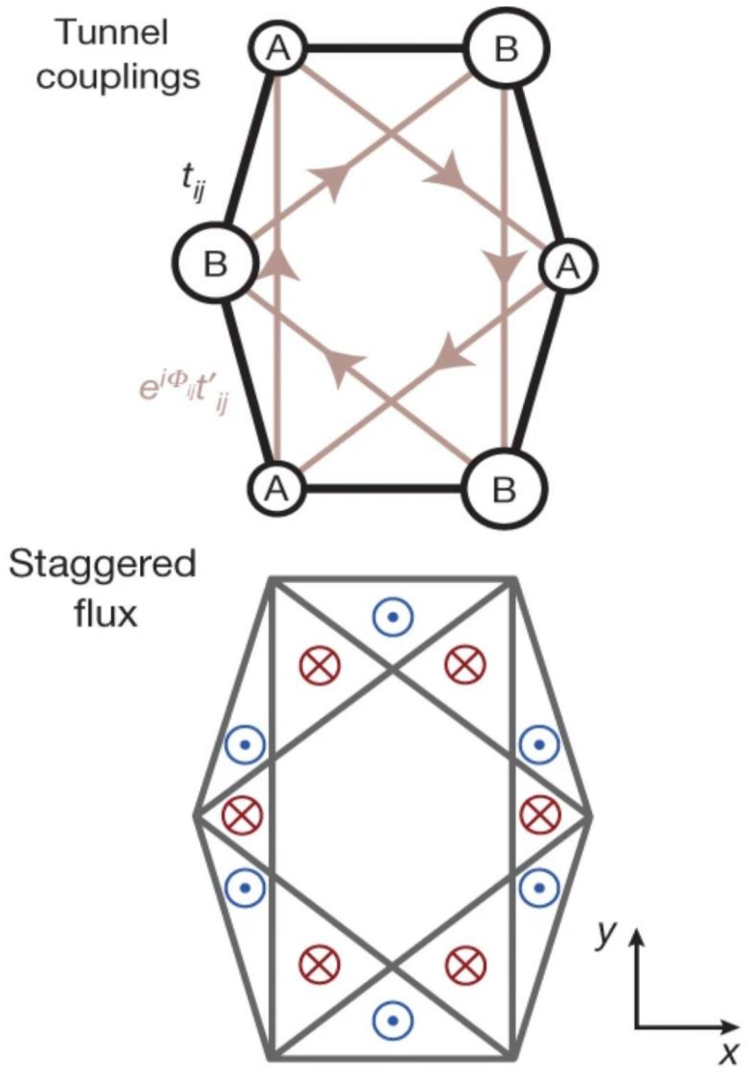

$$H_{NN}(k) = \sum_k \left[ a_k^+, b_k^+ \right] \begin{bmatrix} 0 & H_{12}(k) \\ H_{21}(k) & 0 \end{bmatrix} \begin{bmatrix} a_k \\ b_k \end{bmatrix}$$

$$H_{NNN}(k) = \sum_k \left[ a_k^+, b_k^+ \right] \begin{bmatrix} H_{11}(k) & 0 \\ 0 & H_{22}(k) \end{bmatrix} \begin{bmatrix} a_k \\ b_k \end{bmatrix}$$

$$H(k) = H_{NN}(k) + H_{NNN}(k) + H_{potential}(k)$$

$$H_{potential}(k) = M \sum_k a_k^+ a_k - M \sum_k b_k^+ b_k = M \sum_k [a_k^+, b_k^+] \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a_k \\ b_k \end{bmatrix}$$

# Haldane model




$$H(k) = H_0 I + H_x \sigma_x + H_y \sigma_y + H_z \sigma_z$$

$$E(k)_{\pm} = H_0(k) \pm \sqrt{H_x^2(k) + H_y^2(k) + H_z^2(k)} = H_0(k) \pm |H(k)|$$

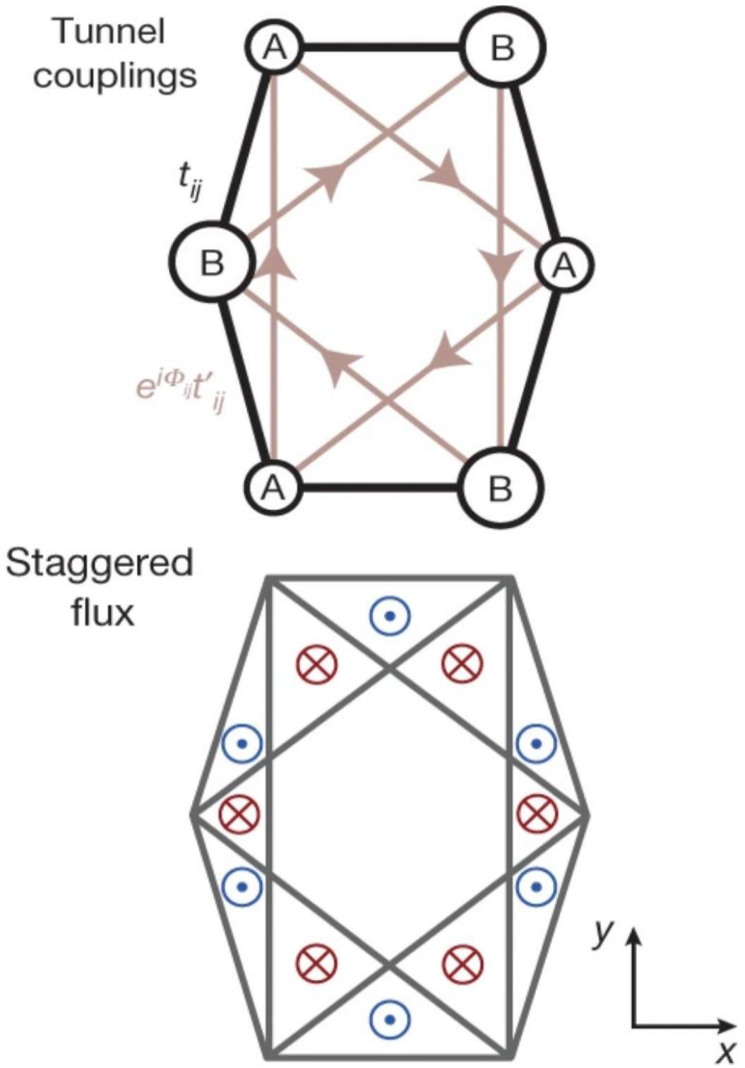
$$H_0 = 2t_2 \cos(\phi) \sum_i \cos(k \cdot v_i)$$

$$H_x = t_1 \sum_i \cos(k \cdot a_i)$$

$$H_y = t_1 \sum_i \sin(k \cdot a_1)$$

$$H_z = [M - 2t_2 \sin(\phi) \sum_i \sin(k \cdot v_i)]$$

# Phase term in NNN hopping



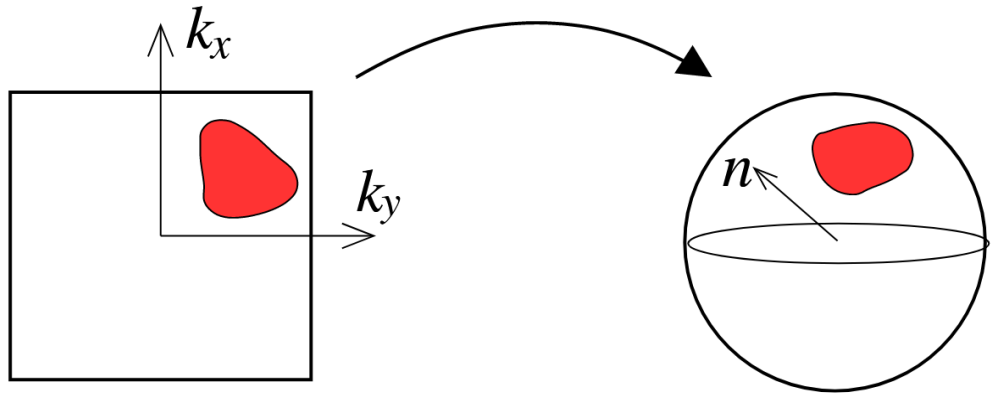

Peierls substitution $t_{ij} \to t_{ij} \exp\left(i\frac{e}{\hbar}\int_i^j \mathbf{A}\cdot d\mathbf{l}\right)$

$$\begin{aligned}
\tilde{H}(t)\tilde{\phi}_{\mathbf{R}}(\mathbf{r}) &= \left[\frac{(\mathbf{p}-q\mathbf{A}(\mathbf{r},t))^2}{2m}+U(\mathbf{r})\right] e^{i\frac{q}{\hbar}\int_{\mathbf{R}}^{\mathbf{r}}\mathbf{A}(\mathbf{r}',t)\cdot d\mathbf{r}'}\phi_{\mathbf{R}}(\mathbf{r}) \\
&= e^{i\frac{q}{\hbar}\int_{\mathbf{R}}^{\mathbf{r}}A(\mathbf{r}',t)\cdot d\mathbf{r}'}\left[\frac{(\mathbf{p}-q\mathbf{A}(\mathbf{r},t)+q\mathbf{A}(\mathbf{r},t))^2}{2m}+U(\mathbf{r})\right]\phi_{\mathbf{R}}(\mathbf{r}) \\
&= e^{i\frac{q}{\hbar}\int_{\mathbf{R}}^{\mathbf{r}}A(\mathbf{r}',t)\cdot d\mathbf{r}'}H\phi_{\mathbf{R}}(\mathbf{r}).
\end{aligned}$$

$$\begin{aligned}
\tilde{t}_{\mathbf{R}\mathbf{R}'}(t) &= -\int d\mathbf{r}\,\tilde{\phi}^*_{\mathbf{R}}(\mathbf{r})\tilde{H}(t)\tilde{\phi}_{\mathbf{R}'}(\mathbf{r}) \\
&= -\int d\mathbf{r}\,\phi^*_{\mathbf{R}}(\mathbf{r})e^{i\frac{q}{\hbar}\left[-\int_{\mathbf{R}}^{\mathbf{r}}\mathbf{A}(\mathbf{r}',t)\cdot d\mathbf{r}'+\int_{\mathbf{R}'}^{\mathbf{r}}\mathbf{A}(\mathbf{r}',t)\cdot d\mathbf{r}'\right]}H\phi_{\mathbf{R}'}(\mathbf{r}) \\
&= -e^{i\frac{q}{\hbar}\int_{\mathbf{R}'}^{\mathbf{R}}\mathbf{A}(\mathbf{r}',t)\cdot d\mathbf{r}'}\int d\mathbf{r}\,\phi^*_{\mathbf{R}}(\mathbf{r})e^{i\frac{q}{\hbar}\Phi_{\mathbf{R}',\mathbf{r},\mathbf{R}}}H\phi_{\mathbf{R}'}(\mathbf{r}),
\end{aligned}$$

# 2D Dirac fermion Hamiltonian

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Brillouin zone Bloch sphere

Sublattice
pseudospin space

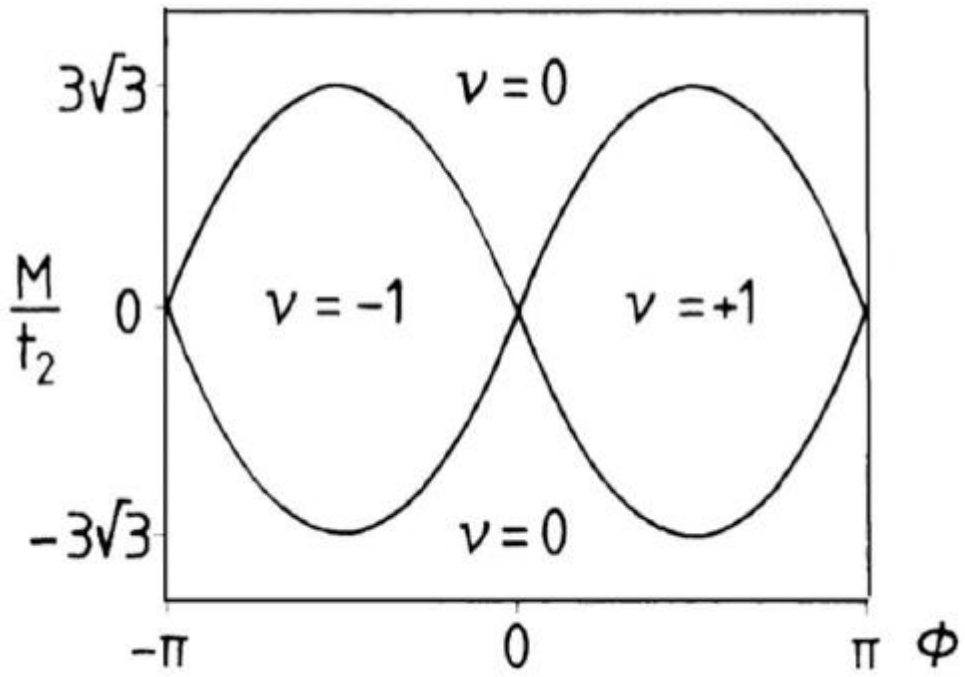
$$\hat{\mathbf{d}}(\mathbf{k}) = \frac{\mathbf{d}(\mathbf{k})}{|\mathbf{d}(\mathbf{k})|} \quad \mathbf{k} \in T^2 \longrightarrow \hat{\mathbf{d}}(\mathbf{k}) \in S^2$$

$$\Omega(\mathbf{k}) = \frac{1}{2}\,\hat{\mathbf{d}} \cdot \left(\partial_{k_x}\hat{\mathbf{d}} \times \partial_{k_y}\hat{\mathbf{d}}\right)$$

$$C = \frac{1}{2\pi}\int_{\mathrm{BZ}} \Omega(\mathbf{k})\, d^2k.$$

$$C = \frac{1}{4\pi}\int_{\mathrm{BZ}} d^2k\; \hat{\mathbf{d}} \cdot \left(\partial_{k_x}\hat{\mathbf{d}} \times \partial_{k_y}\hat{\mathbf{d}}\right)$$

# Chern number




$$H(k) = H_0 I + H_x \sigma_x + H_y \sigma_y + H_z \sigma_z$$

$$H_z = [M - 2t_2 \sin(\phi) \sum_i \sin(k \cdot v_i)]$$

If we have $|M| < |3\sqrt{3} t_2 sin\,(\phi)|$

$$C_{Dirac} = -\frac{1}{2}\, sgn\big(M \pm 3\sqrt{3} t_2\, sin(\phi)\big)$$

$$C_{total} = C_K - C_{K'}$$

TKNN formula $\sigma_{xy} = \frac{e^2}{2\pi\hbar} \sum_\alpha C_\alpha$

# Chern insulator

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A Chern insulator is simply defined as an insulator with a nonzero Chern invariant due to the broken time-reversal symmetry.

$$\mathcal{T} = i\sigma_y K$$

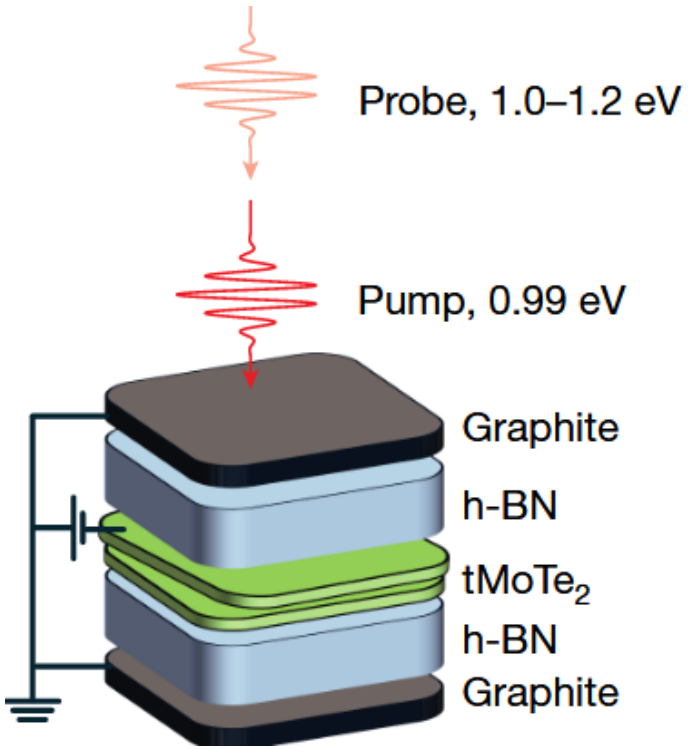
$$\mathcal{T}H(\mathbf{k})\mathcal{T}^{-1} = \sum_i -d_i(\mathbf{k})\sigma_i = -k_x\sigma_x - k_y\sigma_y - m\sigma_z$$

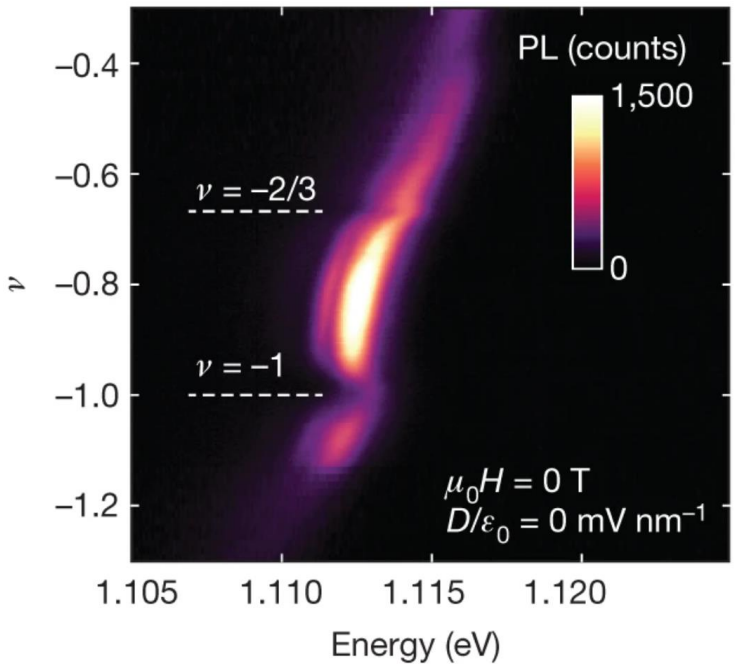
$$\mathcal{T}H(\mathbf{k})\mathcal{T}^{-1} \overset{!}{=} H(-\mathbf{k}) = -k_x\sigma_x - k_y\sigma_y + m\sigma_z$$

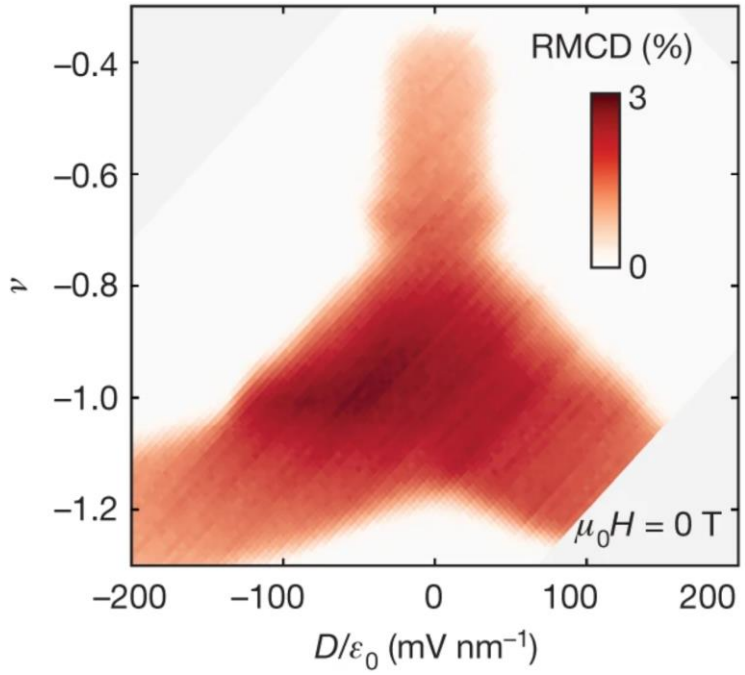
“Topological insulator” is an fermionic insulator with time reversal symmetry and U(1) symmetry (Z2 topological order, C=0).

# Chern insulator in twisted-MoTe2

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Nature 622, 63–68 (2023)

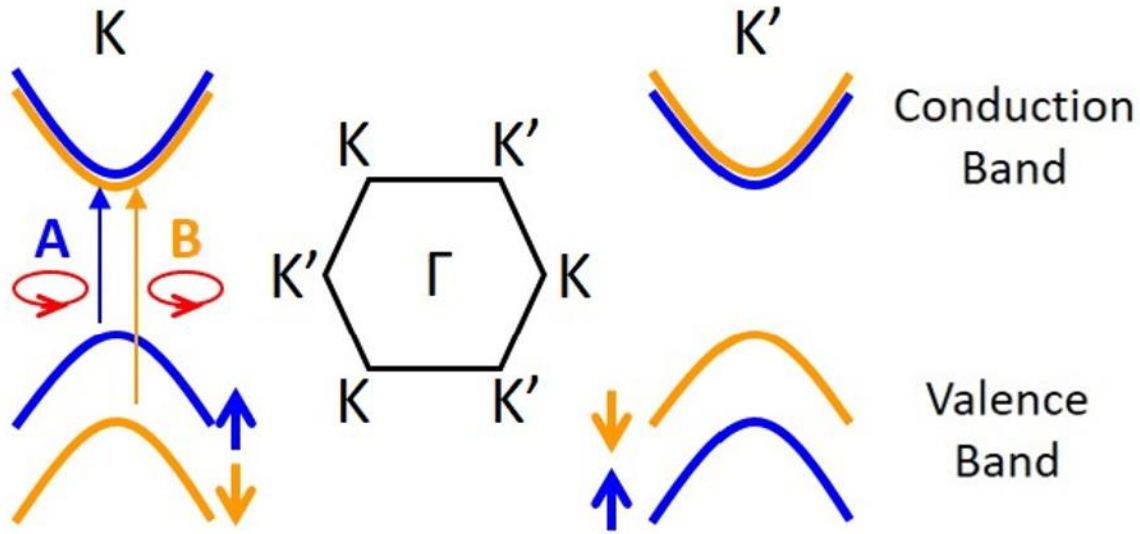
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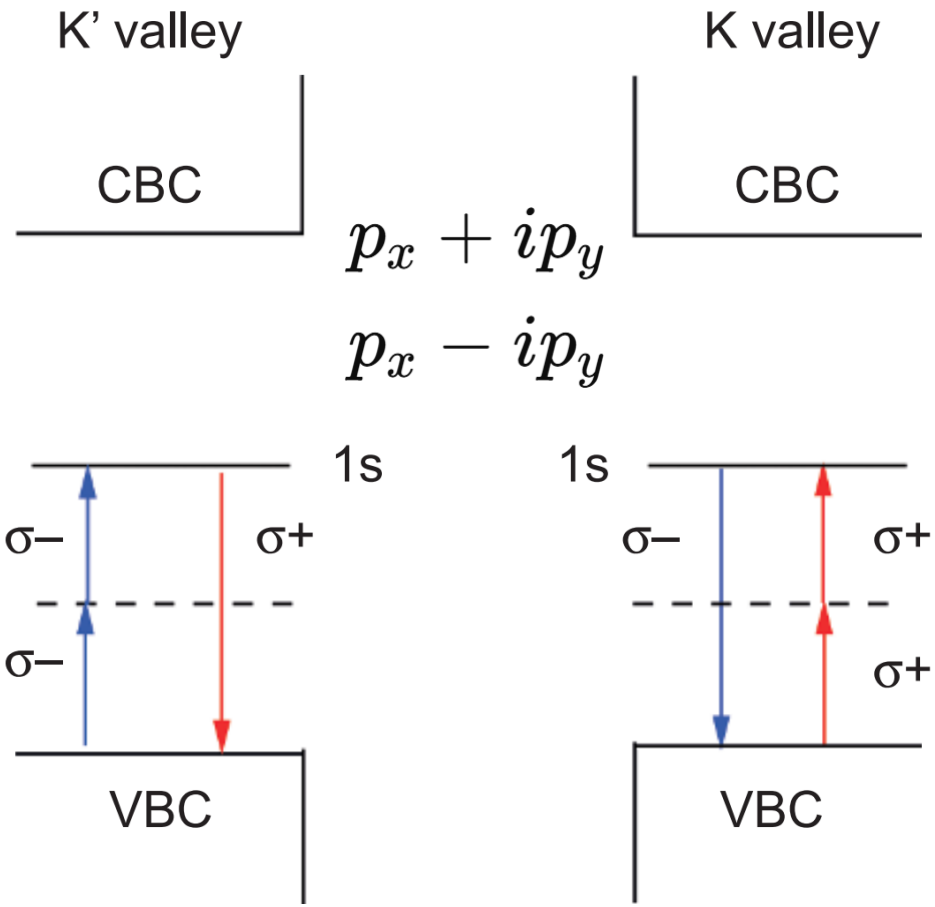
# 03.

# Optical control over Chern number

# Spin-valley Locking and Polarization Selection

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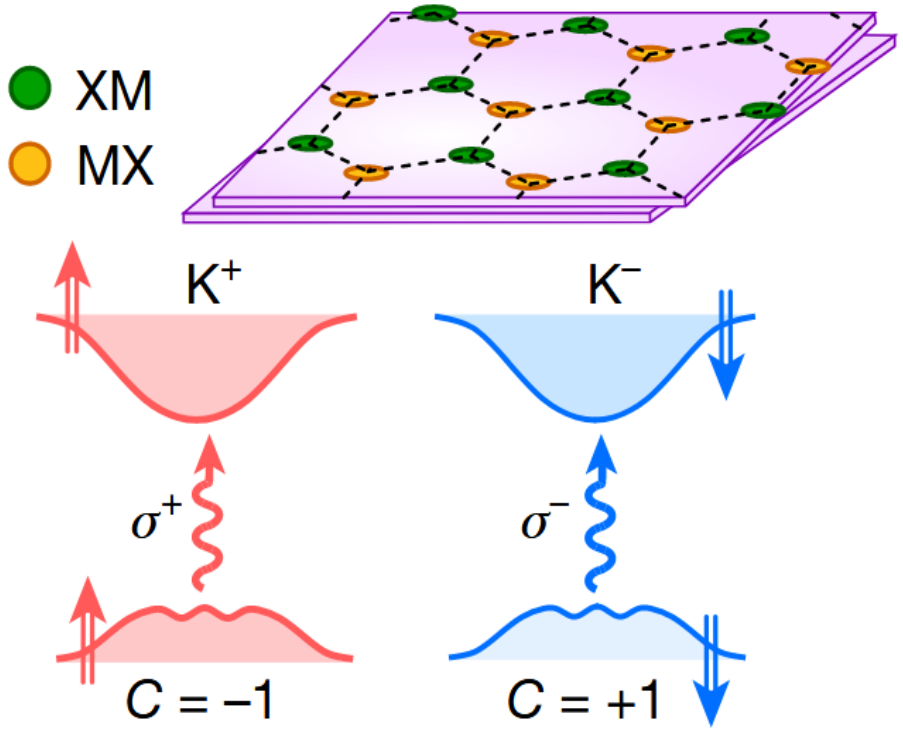



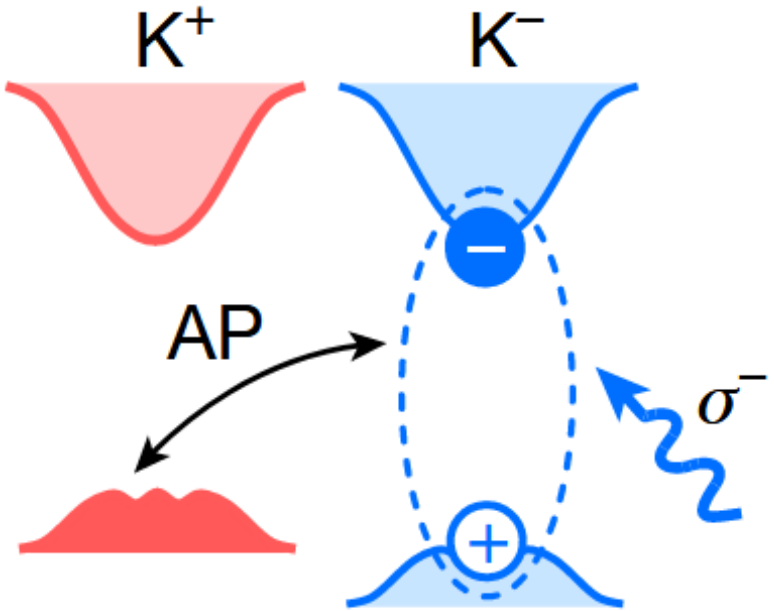

$$M_{\pm}(\mathbf{k}) \propto \langle c, \mathbf{k} | p_x \pm i p_y | v, \mathbf{k} \rangle$$

Light Sci Appl 4, e366 (2015)

# Optical Control over Chern Number

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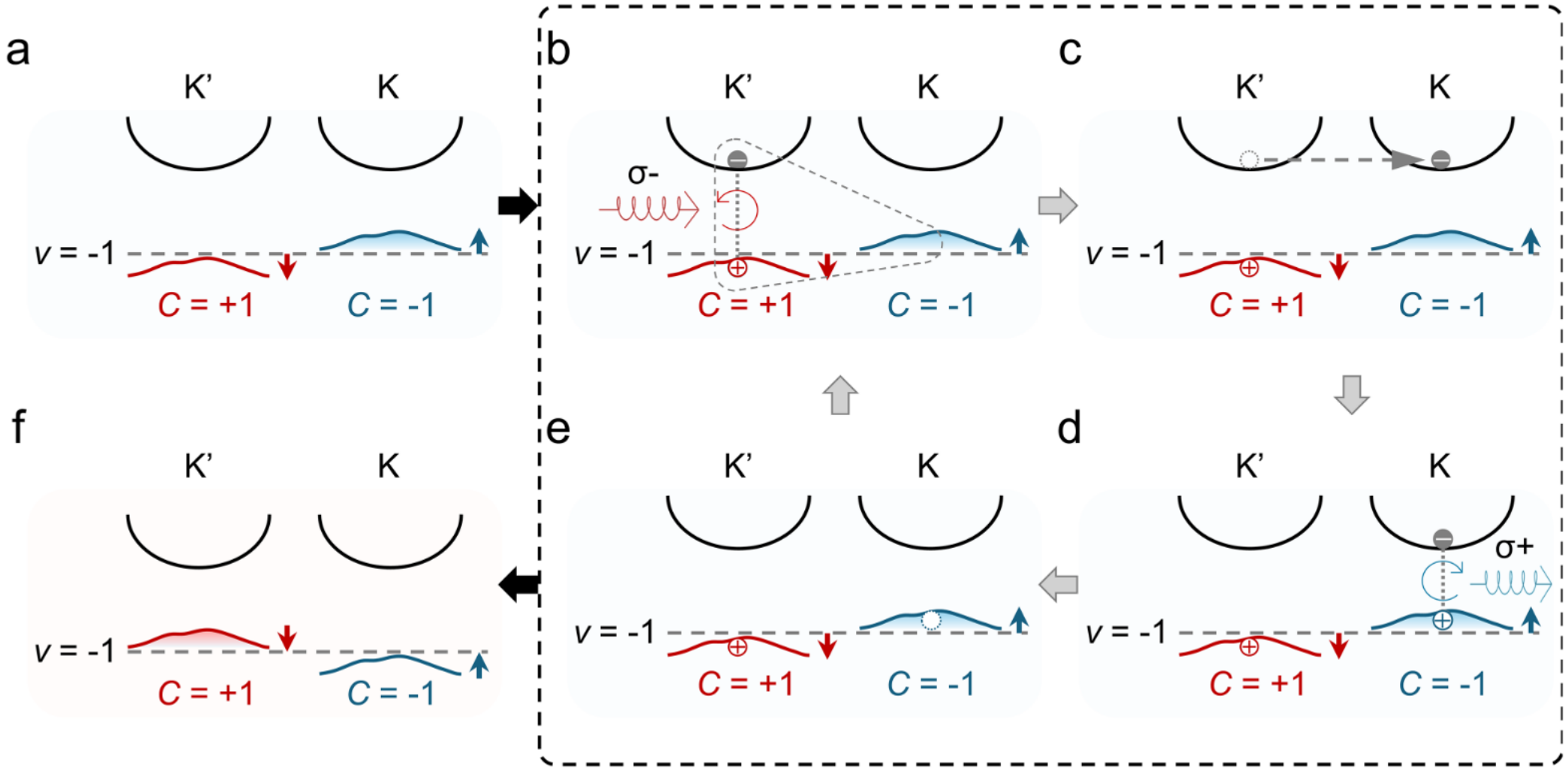



Nature 649, 1153–1158 (2026)

# Mechanisms of Optical Switching

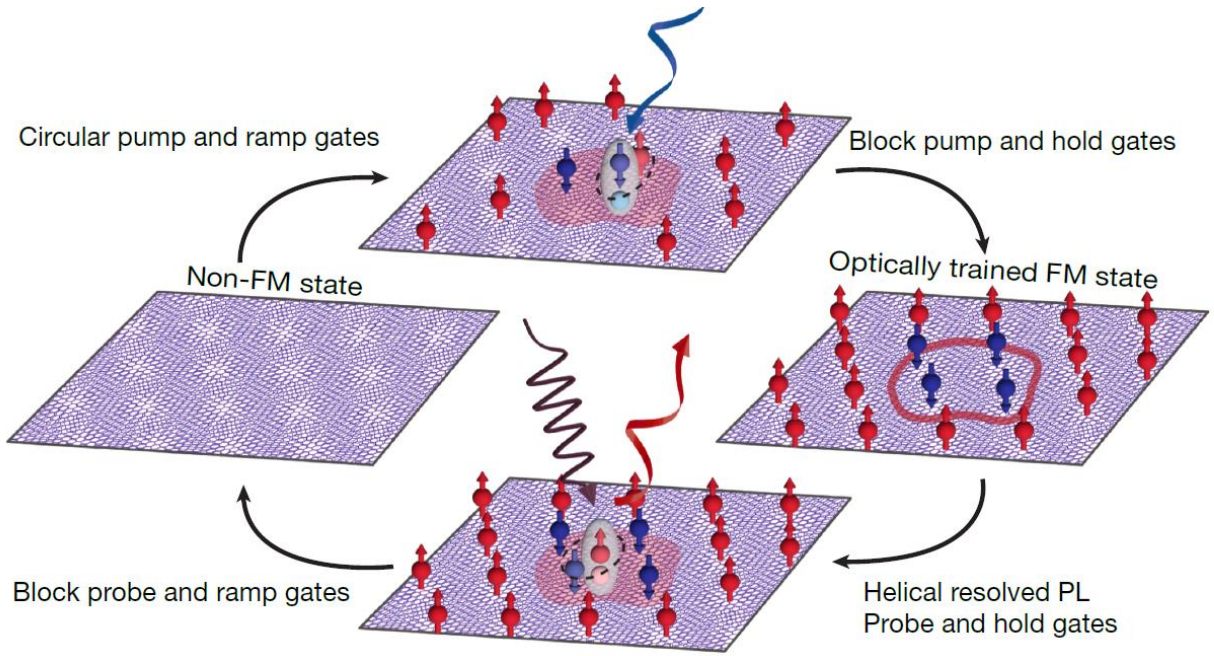
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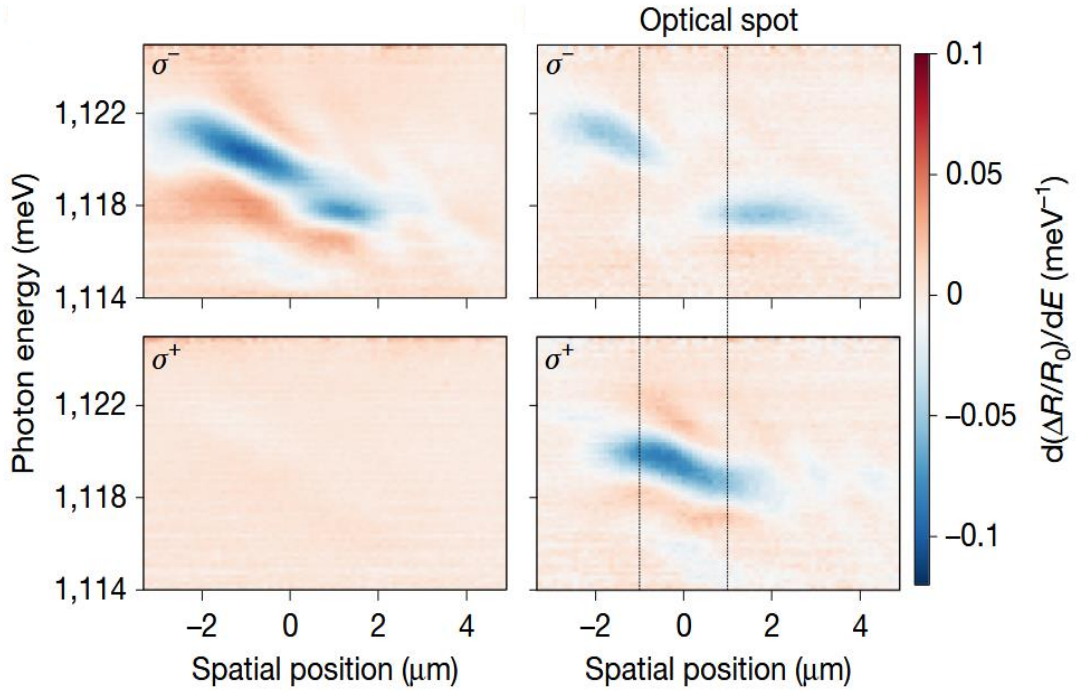



Nature 650, 580–584 (2026)

# Pump cycle

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Nature 649, pages 1147–1152 (2026)



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# THANKS

**Speaker: Zhongqi Xiu**